

Chao Jin

PH.D. CANDIDATE · PEKING UNIVERSITY

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Education

Peking University

PH.D. IN COMPUTER SOFTWARE AND THEORY

- Advisor: Prof. Xin Jin

Beijing, China

Sep. 2023 - Present

Peking University

B.SC. IN COMPUTER SCIENCE AND TECHNOLOGY

- GPA: 3.82/4.0, Ranking: 5/205

Beijing, China

Sep. 2018 - Jun. 2023

Experience

RedNote Dots

RESEARCH INTERN OF LLM TRAINING INFRA

- Mentor: Yue Ban

Beijing, China

Sep. 2025 - Present

StepFun

RESEARCH INTERN OF RL TRAINING INFRA

- Mentor: Yimin Jiang

Beijing, China

Apr. 2025 - Aug. 2025

ByteDance Seed

RESEARCH INTERN OF AI INFRA

- Mentor: Ziheng Jiang, Haibin Lin

Beijing, China

Jul. 2023 - Mar. 2025

Alibaba Cloud

RESEARCH INTERN OF NETWORKED STORAGE

- Mentor: Rui Miao

Beijing, China

Mar. 2022 - Aug. 2022

Publications

ReLibra: Routing-Replay-Guided Load Balancing for MoE Training in Reinforcement Learning

IN PREPRINT.

- Chao Jin, Xinming Wei, Yinmin Zhong, Chengxu Yang, Zili Zhang, Bingyang Wu, Yuliang Liu, Xin Jin

May 2026

MegaScale-MoE: Large-Scale Communication-Efficient Training of Mixture-of-Experts Models in Production

EUROPEAN CONFERENCE ON COMPUTER SYSTEMS (EUROSYS 2026)

- Chao Jin*, Ziheng Jiang*, Zhihao Bai, Zheng Zhong, Juncai Liu, Xiang Li, Ningxin Zheng, Xi Wang, Cong Xie, Qi Huang, Wen Heng, Yiyuan Ma, Wenlei Bao, Size Zheng, Xuegui Zheng, Yanghua Peng, Haibin Lin, Xuanzhe Liu, Xin Jin, Xin Liu
- (* Equal Contribution)

Apr. 2026

Edinburgh, UK

Efficient RL for LLMs with Dynamic and Online Speculative Decoding

THE 1ST FRONTIER AI SYSTEMS WORKSHOP (FAISys 2025)

- Chao Jin, Yinmin Zhong, Zili Zhang, Yimin Jiang, Yibo Zhu

Hong Kong, China

Nov. 2025

MegaScale-Infer: Serving Mixture-of-Experts at Scale with Disaggregated Expert Parallelism

ACM SPECIAL INTEREST GROUP ON DATA COMMUNICATION (SIGCOMM 2025)

- Ruidong Zhu*, Ziheng Jiang*, Chao Jin*, Peng Wu, Cesar A. Stuardo, Dongyang Wang, Xinlei Zhang, Huaping Zhou, Haoran Wei, Yang Cheng, Jianzhe Xiao, Xinyi Zhang, Lingjun Liu, Haibin Lin, Li-Wen Chang, Jianxi Ye, Xiao Yu, Xuanzhe Liu, Xin Jin, Xin Liu
- (* Equal Contribution)

Sep. 2025

Coimbra, Portugal

RAGCache: Efficient Knowledge Caching for Retrieval-Augmented Generation

ACM TRANSACTIONS ON COMPUTER SYSTEMS (TOCS 2025)

- Chao Jin, Zili Zhang, Xuanlin Jiang, Fangyue Liu, Shufan Liu, Xuanzhe Liu, Xin Jin

Sep. 2025

Ditto: Efficient Serverless Analytics with Elastic Parallelism

New York City, NY, U.S.

ACM SPECIAL INTEREST GROUP ON DATA COMMUNICATION (SIGCOMM 2023)

Sep. 2023

- Chao Jin, Zili Zhang, Xingyu Xiang, Songyun Zou, Gang Huang, Xuanzhe Liu, Xin Jin

UltraEP: Unleash MoE Training and Inference on Rack-Scale Nodes with Near-Optimal Load Balancing

IN PREPRINT.

Jun. 2026

- Xinming Wei, Chao Jin, Tuo Dai, Yinmin Zhong, Shan Yu, Chengxu Yang, Bingyang Wu, Zili Zhang, Jing Mai, Qianchao Zhu, Zhouyang Li, Yuliang Liu, Guojie Luo

BigMac: Breaking the Pareto Frontier of Compute and Memory in Multimodal LLM Training

IN PREPRINT.

May 2026

- Zili Zhang*, Chengxu Yang*, Shenglong Zhang, Chenyu Wang, Yufan Zhang, Tuo Dai, Zhouyang Li, Yuhong Ge, Chao Jin, Xin Jin, Yuliang Liu

HydraServe: Minimizing Cold Start Latency for Serverless LLM Serving in Public Clouds

Renton, WA, U.S.

USENIX SYMPOSIUM ON NETWORKED SYSTEMS DESIGN AND IMPLEMENTATION (NSDI 2026)

May 2026

- Chiheng Lou, Sheng Qi, Chao Jin, Dapeng Nie, Haoran Yang, Xuanzhe Liu, Xin Jin

Heddle: A Distributed Orchestration System for Agentic RL Rollout

IN PREPRINT.

Mar. 2026

- Zili Zhang, Yinmin Zhong, Chengxu Yang, Chao Jin, Bingyang Wu, Xinming Wei, Yuliang Liu, Xin Jin

StreamRL: Scalable, Heterogeneous, and Elastic RL for LLMs with Disaggregated Stream Generation

IN PREPRINT.

Apr. 2025

- Yinmin Zhong, Zili Zhang, Xiaoniu Song, Hanpeng Hu, Chao Jin, Bingyang Wu, Nuo Chen, Yukun Chen, Yu Zhou, Changyi Wan, Hongyu Zhou, Yimin Jiang, Yibo Zhu, Daxin Jiang

FaaSPR: Latency-oriented Placement and Routing Optimization for Serverless Workflow Processing

IEEE/ACM TRANSACTIONS ON NETWORKING (TON 2025)

Mar. 2025

- Yunshan Jia, Chao Jin, Qing Li, Xuanzhe Liu, Xin Jin

Pyxix: Scheduling Mixed Tasks in Disaggregated Datacenters

IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS (TPDS 2024)

Jun. 2024

- Sheng Qi, Chao Jin, Mosharaf Chowdhury, Zhenming Liu, Gang Huang, Xuanzhe Liu, Xin Jin

Jolteon: Unleashing the Promise of Serverless for Serverless Workflows

Santa Clara, CA, U.S.

USENIX SYMPOSIUM ON NETWORKED SYSTEMS DESIGN AND IMPLEMENTATION (NSDI 2024)

Apr. 2024

- Zili Zhang, Chao Jin, Xin Jin

Fast, Approximate Vector Queries on Very Large Unstructured Datasets

Boston, MA, U.S.

USENIX SYMPOSIUM ON NETWORKED SYSTEMS DESIGN AND IMPLEMENTATION (NSDI 2023)

Apr. 2023

- Zili Zhang, Chao Jin, Linpeng Tang, Xuanzhe Liu, Xin Jin

Melon: Breaking the Memory Wall for Resource-Efficient On-Device Machine Learning

Portland, OR, U.S.

ACM INTERNATIONAL CONFERENCE ON MOBILE SYSTEMS, APPLICATIONS, AND SERVICES (MOBISYS 2022)

June 2022

- Qipeng Wang, Mengwei Xu, Chao Jin, Xinran Dong, Jinliang Yuan, Gang Huang, Xin Jin, Yunxin Liu, Xuanzhe Liu

Teaching

Operating Systems (Honor Track)

Peking University

TEACHING ASSISTANT

Feb. 2024 - Jun. 2024

Introduction to Computer System

Peking University

TEACHING ASSISTANT

Sep. 2021 - Jan. 2022

Services

IEEE Journal on Selected Areas in Communications (JSAC)

REVIEWER

Feb. 2026

Honors & Awards

UNIVERSITY AWARDS

2024	Merit Student , Peking University Award	<i>Peking University</i>
2024	Huawei Scholarship , Peking University Award	<i>Peking University</i>
2023	Presidential Scholarship , Peking University Award	<i>Peking University</i>
2021	Learning Excellence Award , Peking University Award	<i>Peking University</i>
2020	Merit Student , Peking University Award	<i>Peking University</i>
2020	Xiaomi Scholarship , Peking University Award	<i>Peking University</i>